Wasl at KnowledgeGraphEval 2026: Multi-Head Nested BIO Tagging and Constraint-Aware Relation Classification for Arabic Knowledge Graph Construction

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Abstract

We describe Team Wasl's systems for the KnowledgeGraphEval 2026 shared task at ArabicNLP 2026, which decomposes Arabic knowledge-graph construction into two evaluated stages: nested NER under domain shift and relation extraction between given mentions. For Subtask 1 (AdaptNER), we train a multi-head BIO tagger (one three-way O/B/I head per entity type over a shared pretrained Arabic encoder) on Wojood and combine three random seeds by span-level majority vote. For Subtask 2, forced-choice classification over the 40 Wojood^{Relations} relation types, we fine-tune a typed-marker classifier with type-constraint decoding, ensembling five grouped folds by logit averaging. Our submissions scored 0.7483 micro-F1 on AdaptNER's blind Konooz slice (ten unseen domains), 4th of the eight officially ranked teams and 11.8 points above the official baseline, and 0.9571 micro-F1 on relation extraction with the no-relation class masked at decoding, 2nd at the results release; re-scored by the organizers over all 41 classes, the same ensemble reaches 0.9328, 4th in the organizers' final ranking. We analyze where the 18.6-point domain-shift tax on NER falls: chiefly on recall. We further show that type-constraint decoding and ensembling are partially redundant (+0.9 points on a single RE model, +0.1 on the ensemble) and that the surviving errors share an argument-type signature that no type constraint can separate.

1 Introduction

Knowledge-graph construction from Arabic text requires two capabilities in sequence: finding the entities and typing the relations between them. The KnowledgeGraphEval 2026 shared task (Aljabari et al., 2026) decomposes this pipeline into two subtasks. **AdaptNER** asks for nested named-entity recognition over 21 entity types, trained on the Wojood corpus and evaluated blind on Konooz, ten domains unseen in training. **Relation Extraction** asks for forced-choice classification of a given entity-mention pair into one of 40 relation types from Wojood^{Relations}. We entered both subtasks as Team Wasl and finished 4th and 2nd of the eight officially ranked teams at the results release (§6.1 reports the organizers' later re-scoring of Subtask 2).

Our approach favors disciplined engineering of well-understood components over novel architecture: (1) a faithful reimplementation of the multi-head nested-BIO recipe for AdaptNER; (2) type-constraint decoding for RE, which penalizes relation logits inadmissible for the pair's argument types; (3) ensembling matched to each task's variance source, random seeds with span-level voting for NER and grouped 5-fold logit averaging for RE; and (4) a leakage-free protocol, under which Konooz never touches training, validation, or model selection.

2 Related Work

Our systems assemble published components of the Wojood ecosystem rather than new architecture. Wojood supplies the 21-type nested Arabic NER corpus and the multi-task BIO reference tagger (Jarrar et al., 2022), extended by two WojoodNER shared tasks (Jarrar et al., 2023, 2024), whose 2024 out-of-domain Open Track is AdaptNER's direct precursor. Wojood^{Relations} adds 40 relation types over the same entity scheme (Aljabari et al., 2025b), Wojood^{Hadath} extends the annotation layer to event arguments (Aljabari et al., 2024), *WojoodOntology* places the entity and relation types in an ontology aligned with Schema.org and Wikidata (Aljabari et al., 2025a), and Konooz, the corpus behind the blind test, measures 12–20 F1 points of cross-domain degradation for Arabic NER models (Hamad et al., 2025). From the wider literature we adopt the multi-head per-type BIO formulation with the unified imbalance-aware

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loss of El Mahdaouy et al. (2023), typed punctuation entity markers for relation classification (Zhou and Chen, 2022), and AraBERTv02 (Antoun et al., 2020), the encoder of the organizers' official baselines for both subtasks.

3 Task and Data

Subtask 1: AdaptNER. Training data is Wojood with its standard train/validation split. The blind test set is a slice of Konooz released without labels: 50,175 tokens of MSA across its ten domains. The full corpus also spans sixteen dialects, which the task excludes, so the shift studied here is domain shift within MSA, not dialect shift. Systems submit one IOB2-tagged file and are scored by nested span micro-F1. Task rules prohibit using any part of Konooz, including its public development portion, for training, validation, or model selection.

Subtask 2: Relation Extraction. Wojood^{Relations} provides sentences with a marked subject and object mention; the system outputs exactly one of 40 positive relation types per instance. The task page states that the test set contains no negative pairs, under which micro-F1 equals precision, recall, and accuracy; the official protocol, clarified after the results, scores the full 41-class output, so a `no_relation` prediction counts as a miss (§6.1). The distributed training material contains 17,381 labeled records; the blind test set has 4,386 pairs. Both systems train only on the distributed task data (the Subtask 2 reporting requirement), and all predictions are automatic.

4 System Overview

4.1 AdaptNER

Tagger. We fine-tune `aubmindlab/bert-base-arabertv02` with 21 independent per-type O/B/I heads (a single 21×3-wide linear layer over the shared encoder states), which emit the 21-column submission format directly. Following El Mahdaouy et al. (2023), we train each head with $0.4\,\text{CE}+0.2\,\text{Dice}+0.2\,\text{Tversky}(\alpha{=}0.5)+0.2\,\text{Focal}(\gamma{=}2)$, and the total loss adds a variance penalty across per-head losses to stop frequent types from dominating rare ones; its weight, 5.0, is the value El Mahdaouy et al. fix without tuning, and we did not tune it either (sensitivity: Appendix H). We label only the first sub-token of each word and split over-long sentences at word boundaries within the 512-token budget. Decoding applies standard IOB2 rules with a single repair (an orphan `I-x` starts a new span).

Seed ensemble. The submitted ensemble trains three seeds (13, 42, 77) on train+validation, with a fixed 29-epoch schedule, the train-only model's validation peak; folding validation into training forfeits early stopping. We combine the seeds in three ways: **mean-prob** (argmax of the seed-averaged softmax), **union** (any span predicted by $\geq$1 seed, a recall lever), and **majority vote** ($\geq$2 seeds, a precision lever); overlap resolution: Appendix C.

4.2 Relation Extraction

Classifier. We pair the same encoder with typed punctuation entity markers in the style of Zhou and Chen (2022): the subject mention is wrapped as `@ * type * mention @` and the object as `# ^ type ^ mention #`, so the argument types enter as plain text with no new vocabulary; a two-layer feed-forward head over the concatenated start-marker states classifies into 41 classes (40 relations + `no_relation`). We train on the full 41-way problem. Our official submission set the `no_relation` logit to $-\infty$ at decoding, following the task page's statement that the test set holds no negatives; at the organizers' request after the results we resubmitted the same ensemble with the mask removed (§6.1).

Argument typing at test time. The blind test provides mentions but not the entity types the markers and constraints need. We join each record's sentence id to Wojood's gold sentences (entity annotations only, never relation labels) and match the mention string to a gold entity surface, falling back to a corpus-wide mention→type lexicon and finally to UNK, which disables that instance's constraints rather than guessing (coverage: Appendix C).

Constraint decoding. We build a table of inadmissible relations per (subject type, object type) pair from the task's relation type specification unioned with every type pair observed in training, since gold annotations occasionally violate the specification. At decode we subtract a fixed constant from each inadmissible relation's logit, a soft penalty rather than a hard mask; a sweep over penalties 0–4 on development data selected 3.0.

Fold ensemble. The final system averages the raw test logits of five models trained on grouped

Task	System	Mi-F1	P	R	Ma-F1
NER	v1 single s13	0.7385	0.7874	0.6953	0.6249
NER	v2 vote ens.	**0.7483**	0.8060	0.6982	0.6316
RE	v1 single+p3	0.9519	—	—	—
RE	v2 5-fold+p3	**0.9571**	—	—	—
RE	v2 unmasked†	0.9328	0.9716	0.8969	—

Table 1: Official blind-test results. RE is scored forced-choice single-label, so P and R equal micro-F1 for the masked rows; no macro-F1 is published. †the same ensemble decoded without the `no_relation` mask, scored by the organizers over all 41 classes after the results release (§6.1).

5-fold splits, grouped by sentence id because sentences average 2.75 triples and row-level splits would leak; each fold early-stops on its own held-out fold. The pooled out-of-fold (OOF) predictions re-validate the penalty choice on all 17,381 records.

5 Experimental Setup

All experiments ran on a single NVIDIA T4 (free Kaggle tier) with deterministic seeds; no reported system cost more than $\approx$4 GPU-hours in aggregate. Appendix A lists the hyperparameters; code is available at `https://github.com/Salah-Sal/kgeval2026-team-rabt`.

Validation protocol. AdaptNER model selection used only the Wojood validation split (early stopping on span micro-F1); we scored the Wojood test split once per finished system as an in-domain reference; Konooz never entered any loop. For RE, we made development decisions on a single grouped held-out split (15,207 train / 2,174 dev), replaced for the final system by grouped 5-fold cross-validation.

We used two submission slots per track: v1 single models and v2 ensembles (Table 5).

6 Results

6.1 Official test results

Among the eight officially ranked teams we finished **4th** on AdaptNER, +0.12 micro / +0.19 macro over the no-adaptation AraBERTv02 baseline (0.63 / 0.44), and **2nd** on relation extraction at the results release (0.9571, behind 0.9603; +0.026 over the organizers' baseline retrained on the released split, 0.9309). After the release the organizers re-scored Subtask 2 over all 41 classes, and the same ensemble, decoded without the `no_relation` mask, scores 0.9328 (P 0.9716, R 0.8969; 4th of the seven ranked teams in the organizers' final table): the scorer counts each of the 337 `no_relation` predictions as a miss, and 264 of them had been correct under the mask. Ensembling improved both tracks on the blind test (Table 1): +0.98 micro-F1 for AdaptNER and +0.52 for RE under the masked protocol, where the pooled-OOF estimate anticipated the blind score to within 0.3 points.

6.2 Development results and analysis

Ensembling where the variance is. In domain, majority vote is the strongest seed combination: +0.6 over the ensemble's own seed-13 member (train+val, 0.9268; Table 3), the reference for all in-domain ensemble deltas; the union, the intended recall lever, buys +0.9 recall for -2.0 precision. On the blind set the three seeds largely agree (the union holds only 8% more spans than the largest single seed), so the recall deficit under shift is *systematic*: entities no seed finds. We therefore submitted the precision-preserving vote, and the blind result bore the reading out: +1.9 precision, +0.3 recall over the submitted single model (v1).

The domain-shift tax falls on recall. The submitted single model (v1: train only, early-stopped seed 13) drops 18.6 points from its in-domain Wojood test score (0.9240) to blind Konooz (0.7385), a domain shift within MSA, and the drop falls chiefly on recall (-23.6 points, vs. -13.0 precision), reversing the in-domain P–R profile (Table 3). Two anchors put 0.7385 in context: 10.9 points above the official no-adaptation baseline, and at the top of the 12–20-point degradation band Konooz itself measures (Hamad et al., 2025).

Constraint decoding and ensembling are partially redundant. The type-constraint penalty is worth +0.92 points on the single model but only +0.08 pooled-OOF on top of the 5-fold ensemble (Table 4): most type-inadmissible errors are single-model flukes that logit averaging already removes. The penalty's remaining value concentrates in macro-F1 (+1.0 point OOF), i.e., in rare relations, which is why we selected the penalty on macro-F1 rather than on the official micro-F1: micro-F1 peaks at 4.0, by 0.0001, and the whole sweep spans 0.0008 (Table 6). The recipe: ensemble first, then constrain.

6.3 Error analysis

RE: the surviving errors share type signatures. We reran one fold model of the submitted ensemble on its held-out fold (3,476 records), leaving 74 misclassified positives, and their structure explains why the constraint penalty saturates: the residual confusions hold between relations with the *same* argument-type signature, which no type constraint can separate by construction. Two of the largest clusters are PERS–GPE `lives_in` vs. `president_of` (11) and GPE–GPE `capital_of` vs. `geopolitical_division` (7). Two patterns recur. First, a pull toward the type pair's majority relation even against an explicit lexical cue: in برنامج القدس الفلسطينية عاصمتنا ("the program 'Palestinian Jerusalem, Our Capital' "), gold `capital_of` is predicted as `geopolitical_division`, which outnumbers it 59-to-1 in training. Second, genuine overlap between readings: for رئيسة وزراء بريطانيا العظمى، مارغريت تاتشر ("the Prime Minister of Great Britain, Margaret Thatcher"), the model outputs `president_of`, the closest leadership relation the inventory offers, while gold encodes `lives_in`; the confusion runs both ways (6 vs. 5 cases), unlike the one-way majority pull above. On the blind test the penalty flips only 6 of 4,386 ensemble predictions.

NER: metonymy, attachment, and register. On Wojood test (0.9240), errors partition into four mutually exclusive categories (Appendix C). Exact-boundary type confusions number only 138, and 60% of them are ORG↔GPE metonymy: in أنتجت بولندا بنشاط البنادق الهجومية ("Poland actively produced assault rifles"), بولندا is gold-annotated ORG as a state actor but predicted GPE, its default type. Same-type boundary errors (510) concentrate on trailing locatives and appositions, e.g. gold نافورة تابيثا ("the Tabitha Fountain") predicted with في يافا ("in Jaffa") attached. Outright misses (591) and spurious spans (876) both skew toward NORP, ORG, and EVENT, and misses reach into colloquial-register occupations such as شاويش ("sergeant").

Blind-set behavior. Without blind gold, we can characterize out-of-domain errors only behaviorally. Seed unanimity over predicted spans orders the ten domains by distance from the training register: spans predicted by a single seed account for 13–15% of all predicted spans in Finance and Economics, the registers closest to Wojood's newsheavy composition, but 28% in Science and Sport (Table 7).

6.4 Post-submission loss ablation

After the submission close we ablated the unified loss (Appendix H; no run touched Konooz or a scored submission). The result is negative: plain cross-entropy slightly outperforms the submitted recipe in domain (Table 8; single-seed runs), and partial mixtures collapse rare types. The variance penalty functions as a guard against the instability the other loss terms introduce (+12.4 macro over the penalty-free mixture), not as a lever over CE.

7 Conclusion

Disciplined engineering of published components produced 4th of 8 on nested NER under domain shift and 2nd of 8 on relation extraction at the results release, 4th of 7 after the organizers' 41-class re-score: one Arabic encoder, two task-shaped heads, ensembling matched to each task's variance, and a leakage-free protocol. The clearest finding is negative space: the domain-shift tax falls on recall and is shared across seeds, so closing it requires target-domain adaptation, not better combination of Wojood-only models (design alternatives: Appendix G).

8 Limitations

The per-type BIO-head architecture cannot represent nesting of two same-type entities; Konooz happens to contain none, so this costs nothing here, but the architecture is not a general nested-NER solution. We used no external data, augmentation, or domain-adaptive pretraining, so our numbers are a floor for this architecture family; each task also relies on a single encoder family. With no labeled out-of-domain validation permitted, the choice among ensemble variants rested on in-domain scores and the seed-agreement analysis alone, and the rejected variants remain unmeasured on the blind set. Development metrics use our own scorer implementations, reconciled against submission rehearsals but not the official evaluation code; only Codabench numbers are authoritative. RE argument typing at test time is a heuristic join against Wojood gold entities; 96 subject and 91 object sides could not be typed and ran with untyped markers and constraints disabled. No result carries a confidence interval or significance test. Seed variance on the blind set is unmea-

sured, since each seed was submitted only inside the ensemble, so the +0.98 ensemble gain there is a single-seed comparison; in domain, the three seeds span 0.20 points (Table 3).

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A Hyperparameters

	AdaptNER	**RE**
Encoder	arabertv02	arabertv02
Batch / max length	16 / 512	32 / 256
Learning rate	1e-5, γ=0.95	2e-5
Weight decay	1e-3	1e-2
Epochs	$\leq$40, pat. 5[a]	$\leq$10, pat. 2
Dropout / grad clip	0.1 / 1.0	0.1 / 1.0
Seeds	13[b]	13[c]

Table 2: Hyperparameters. Both tasks: AdamW (no decay on bias/LayerNorm), fp16 autocast with gradient scaling. [a]fixed 29 epochs for the ensemble; [b]13/42/77 for the ensemble; [c]13+fold for the ensemble. AdaptNER LR decays exponentially per epoch.

B Development Results

Model	**P**	**R**	**Mi-F1**
v1 single (train only, s13)[a]	0.9169	0.9313	0.9240
Ens. member s13[b]	0.9183	0.9353	0.9268
Ens. member s42	0.9143	0.9356	0.9248
Ens. member s77	0.9147	0.9362	0.9253
Mean-probability	0.9211	0.9379	0.9295
Span union ($\geq$1)	0.8981	0.9440	0.9205
Majority vote ($\geq$2)	0.9281	0.9377	**0.9329**

Table 3: Wojood-test (in-domain) span micro P/R/F1. [a]the submitted single model, early-stopped on Wojood-val (0.9208); it is the reference for the 18.6-point domain-shift drop. [b]the reference for the ensemble deltas in §6.2; ensemble members train on train+val for a fixed 29 epochs and cannot report a validation score since the validation split is folded into training.

Config	**Eval**	**Mi-F1**	**Ma-F1**
Single, no constraints	held-out	0.9378	0.656
+ penalty 3.0	held-out	0.9470	0.705
5-fold ens., no constr.	pooled OOF	0.9592	0.638
+ penalty 3.0	pooled OOF	0.95995	0.648

Table 4: RE development results (micro-F1, macro-F1), positives only. Per-fold best validation micro-F1 spans 0.9518–0.9670 (mean 0.9590).

C Prediction Statistics

- RE marker placement: when a mention string occurs more than once in its sentence (5.9% of instances), we mark the disjoint occurrence pair with minimal distance; when the object is embedded inside the subject mention ($\approx$20%), markers are nested so both stay in context.

- AdaptNER seed combination: for union and vote, same-type overlaps resolve greedily by (votes, length, position).

- AdaptNER blind test: the v1 single model predicts 5,780 spans over 50,175 tokens (density 11.5%, close to Wojood's), all 21 types emitted, 201 BIO-sequence repairs during decoding; the submitted vote ensemble predicts 5,670 spans (Table 7). Health yields the fewest predictions in both (228).

- RE blind test: the ensemble changed 97/4,386 predictions (2.2%) vs. the single model, concentrated in the `located_in`/`subsidiary` and `president_of`/`lives_in` confusion clusters, the same affiliation/location clusters that dominate development-set confusion matrices. 34/40 relation types appear in the ensemble output; `located_in` alone accounts for 46%.

- RE test-time typing: the sentence-id join located the source sentence for 4,360/4,386 records (99.4%); with the lexicon fallback, 4,290 subjects (97.8%) and 4,295 objects (97.9%) received a type; the remaining 96 subjects and 91 objects were typed as unknown, disabling their constraints rather than guessing.

- NER error taxonomy (§6.3): each errant gold span is paired with at most one errant predicted span, by precedence: exact boundaries with a different type (confusion), then same type with overlapping boundaries (boundary). Unpaired gold spans count as misses, unpaired predicted spans as spurious, so the four categories are mutually exclusive by construction.

D Submission Manifest

Track	System	Date	Score
AdaptNER	v1 single s13	Jul 22	0.7385
AdaptNER	v2 3-seed vote	Jul 24	0.7483
RE	v1 single + p3.0	Jul 22	0.9519
RE	v2 5-fold + p3.0	Jul 24	0.9571
RE	v2 unmasked[a]	Sep 4	0.9328

Table 5: Officially scored Codabench submissions (2026; micro-F1). Full configs and data hashes are recorded per run in the project's experiment ledger. [a]41-class re-score at the organizers' request after the results release; same ensemble logits, `no_relation` mask removed.

E Constraint-Penalty Sweep

Penalty	0	1	2	3	4
micro-F1	.95920	.95952	.95984	.95995	.96006
macro-F1	.638	.645	.646	**.648**	.647

Table 6: Pooled out-of-fold penalty sweep for the 5-fold RE ensemble (all 17,381 records, positives only). Micro-F1 is monotone and nearly flat, peaking at 4; penalty 3.0 maximizes macro-F1 and is used for all submissions.

F AdaptNER Blind Predictions by Domain

Domain	Spans (vote ens.)
Agriculture	370
Art	459
Economics	667
Finance	690
Health	228
History	660
Law	635
Politics	697
Science	423
Sport	841
Total	5,670

Table 7: Predicted span counts of the submitted vote ensemble on the blind Konooz test slice, per domain (no gold available pre-deadline).

G Design Alternatives Considered

The design phase rejected the following on published evidence rather than taste: LLM in-context NER (published Arabic results of 0.29–0.74 F1), MRC-style per-type querying (21× inference cost), grid decoders whose extra expressiveness the submission format cannot carry, LLM-only RE (the task's GPT-4o RAG baseline trails its supervised one (Aljabari et al., 2025b)), negative downsampling (reported to cost ≈10 F1 in comparable settings), and hard constraint masking (gold annotations occasionally violate the type specification, so constraints must stay soft). Our design notes projected ≈0.73 micro-F1 for zero-shot Wojood→Konooz transfer; the submitted single model scored 0.7385. The adaptation levers designed but not built by the deadline (continued pretraining on the unlabeled test tokens, guarded self-training, domain-targeted augmentation) were projected to be worth a further 3–8 points and are the obvious next step.

H Post-Submission Loss Ablation and Penalty-Weight Sweep

Run August 3, after the July 29 submission close, on the AdaptNER baseline configuration (Wojood-only, seed 13); no ablation run touched Konooz, the leaderboard, or any submitted system. Adding Dice to CE collapses PRODUCT to 0 F1; the full mixture without the variance penalty additionally zeroes UNIT and QUANTITY; the variance penalty rescues all three. Whether E1's slight in-domain edge holds under domain shift is untestable without Konooz gold. A sweep of the penalty weight (P1, P2, P10; run September 4 after the reviews, same setup) shows macro-F1 rising monotonically from 0 to 5.0 and falling back at 10.0, while micro-F1 varies by at most 0.1 points from 0 to 5.0 and drops 0.4 at 10.0; on this single seed the untuned 5.0 is the best of the five weights on macro-F1 and no worse than any other on micro-F1.

	Loss	Val	Test
E1	CE only	.9227 / .8334	.9267 / .8489
E2	+Dice	.9233 / .7777	.9258 / .7674
E3	+Tversky	.9224 / .7780	.9263 / .7863
E4	+Focal (full mix)	.9202 / .7201	.9240 / .7226
P1	full + pen. 1.0‡	.9198 / .7326	.9239 / .7265
P2	full + pen. 2.0‡	.9207 / .7532	.9239 / .7608
E5	full + pen. 5.0†	.9208 / .8324	.9240 / .8462
P10	full + pen. 10.0‡	.9166 / .7711	.9203 / .7941

Table 8: Post-submission loss ablation (run August 3, after the July 29 close) and penalty-weight sweep (run September 4); never part of a scored submission. Wojood val/test span micro-F1 / macro-F1, seed 13, baseline configuration. Mixture weights over (CE, Dice, Tversky, Focal): E2 (.5,.5,0,0), E3 (.4,.3,.3,0), E4/E5/P* (.4,.2,.2,.2); variance-penalty weight as listed, none in E1–E4. †E5 is the submitted recipe. ‡sweep rows, ordered with E4 and E5 along the weight axis.